Lecture plan · for the lecturer

Who a detector misses

University, vision 2

Time
45 min

Materials
A phone with Darsi; Paper for one hand-worked step; The slides from the kit

Learning goals

- Stack convolution, ReLU and pooling.
- Score drawings against a prototype.
- Report the miss rate for each group.

Activity: Read the scores

1. Warm-up (5 min): Draw convolution, then ReLU, then pooling, and ask what each part throws away.
2. Lab (15 min): Pairs read the three scores and the line, and label each result a hit, a miss or a correct rejection.
3. Numbers (10 min): Students work out the miss rate for close-set and for wide faces, and compare it with “right on two of three”.
4. Rule (10 min): The class writes the lab rule together: drawn patterns only, never classmates’ photos.
5. Exit (5 min): Each student names the group the detector missed, with its score.

Discussion

- Why is the training shape not a test?

Exit ticket

- Name the group the detector missed.

Try it in your next lecture
Add one sentence to the lab sheet: photographs of people are out of scope.

Seminar sheet · for students

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Name _____ Group _____ Date _____

1 **The tiny detector was trained on close-set eyes. A wider drawn face scores…**

◯ The same as the box ◯ Below the line ◯ Perfect

2 **A high score on the training shape is enough to trust a detector on new people.**

◯ True ◯ False

3 **Match the picture.**

Close-set drawing	• • • •	A face the detector misses
Wide drawing	• • • •	Correctly not a face
Box	• • • •	In the training average

4 **Order a fair check of a detector.**

☐ Test on shapes it did not train on

☐ Report who it misses

☐ Train on one set

Answers · for the lecturer

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1 **Below the line**

It never saw that spacing, so the match falls under the line.

2 **False**

The wide face is still a face, and the detector missed it.

3 **Close-set drawing — In the training average**
Wide drawing — A face the detector misses
Box — Correctly not a face

Same idea, different spacing, different score.

4 **Top to bottom: 2, 3, 1**

The miss is invisible if you only rescore the training pictures.

For you to think about

What will you do if the class wants to test on their own photos?